Principles OF Testing:

Testing Shows the Presence OF defects, not their absence:

Testing can find defects, but it can never prove that there are no defects left. The goal is to reduce the number OF defects, not to achive complete bug-Free software.

Exchaustive testing is impossible:

It's not Feasible to test every possible input and combination OF conditions for even simple software. testers MUST use risk analysis and prioritize efforts to focus on the MOST critical areas.

Early testing Saves time and money:

Finding and fixing defects early in the development life cycle is significantly cheaper and faster than finding them later static and dynamic testing efforts on those high-risk areas.

The Pesticide Paradox:

If you keep running the same set tests, they will eventually become ineffective at finding new defects. To over come this tests, need to be updated and new different tests must be created regularly.

Defects cluster together:

Most defects are often found in a small number of modules within the software this Priciple also known as the 80/20 rule, suggests focusing testing efforts on those high-risk areas.

Testing is context dependent:

The right testing strategy depends on the context of the software being tested. For example the testing required for an e-commerce site will differ from the testing for a simple mobile game

The absence of errors Fallacy:

Finding and fixing many defects does not guarantee that the software is useful. If the system built was the wrong one and doesn't meet the user's needs, than finding no defects in it is irrelevent.